

The Bratu Problem

When the Right Answer Is “There Isn’t One”

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SIRPY · Problem 14 — The Integral That Wasn’t There

An ongoing series. We are building a public library of problems SIRPY has solved — each entry is one problem, the exact input we handed the solver, the exact output it returned, and an honest read of what happened. No slides. No hand-waving. Numbers you can check.

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Every entry in this library so far has asked the same question: can you solve it, and can you prove the answer is right?

This one asks something different. **Can you prove there is no answer at all?**

That question is harder, and it is the one that matters most in practice. When a solver fails, you cannot usually tell which of two things happened: the method wasn’t good enough, or the problem has no solution. Those demand opposite responses — try harder, or stop and re-examine your model — and almost no tool tells you which one you’re in.

The problem

$$y'' = -\lambda e^y, \quad y(0) = 0, \quad y(1) = 0$$

This is the **Bratu problem**, the standard model for thermal ignition: y is temperature, and the exponential term is heat released by a reaction that accelerates as it gets hotter. λ measures how strongly the reaction feeds back on itself.

The behaviour is famous and physically meaningful. Below a critical value λ_c , the problem has **two** solutions — a cool, stable steady state and a hot, unstable one. As λ rises the two approach each other. At λ_c they merge. **Above** λ_c there is no steady state at all — the physical meaning being thermal runaway: ignition.

That critical value is known in closed form:

$\lambda_c = 3.513830719\dots$

We ran four values of `lam` across it, changing nothing else.

What we handed the solver

```
for lam in (3.5135, 3.5138, 3.513830719, 3.52):
    r = solve_bvp(order=2, f_str=f"-{lam}*exp(y)", x0=0, x1=1,
                  left_bc={0: 0.0, 1: "gamma"}, right_bc={0: 0})
```

A loop. No tuning, no guesses, no per-case handling — and nothing telling SIRPY that a critical value exists or where it is.

$\lambda = 3.5135$ **and** 3.5138 — **below critical**

Both solved. SIRPY noted that a single series could not be trusted across the full interval and switched to a segmented march:

```
INFO: verification by substitution - VERIFIED: satisfies the equation
      to 6.38e-10 on [0, 1] (the equation's scale reaches 11.7; max
      RELATIVE residual, taken pointwise, is 6.19e-11), and the
      boundary data to 4.44e-16. Interface continuity across 5 node(s):
      max one-sided jump in (y..y1) = 1.6e-11 - measured, not assumed.
```

Relative residuals of 6×10^{-11} and 8×10^{-11} , boundary data to 10^{-16} , and the seams between segments measured rather than presumed.

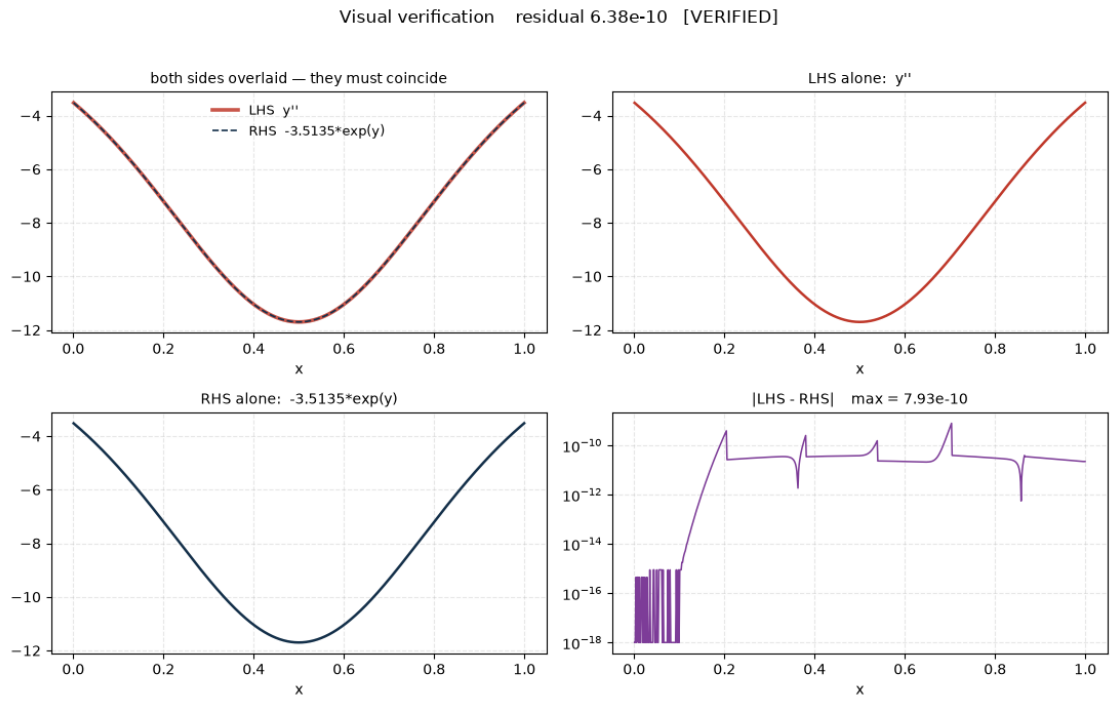
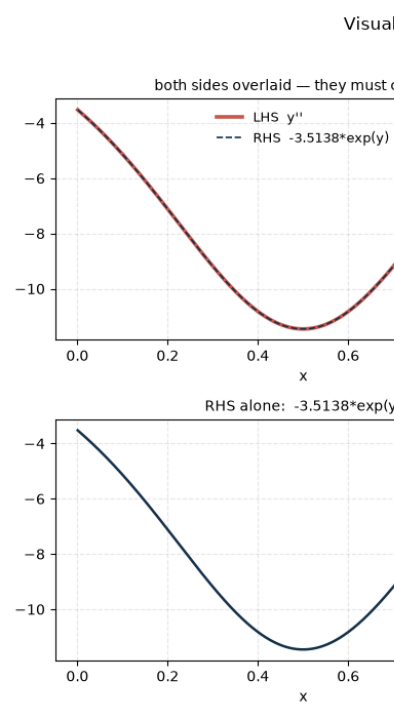
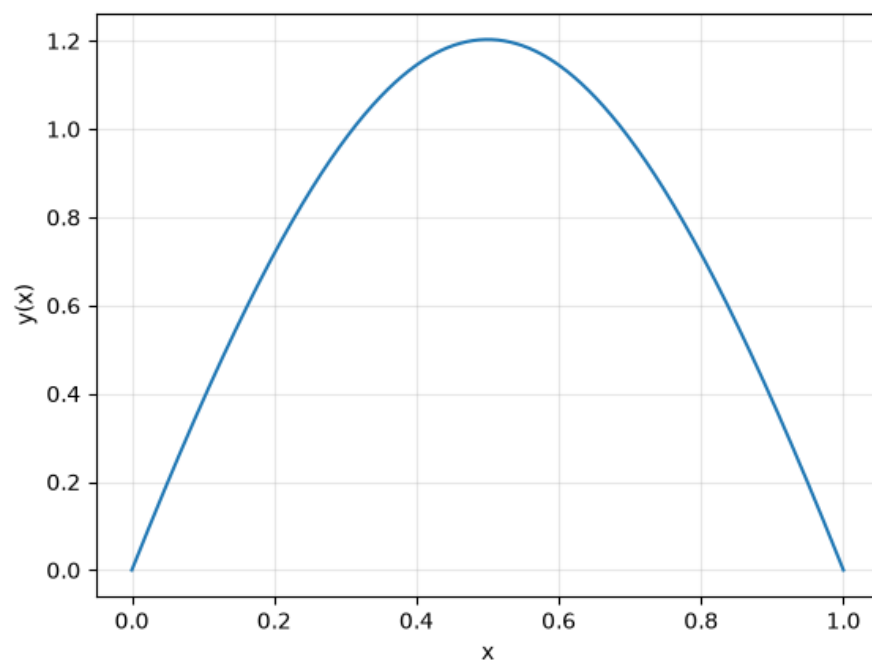


Figure 1: Verification $\lambda = 3.5135$: both sides of the equation compared across the interval.



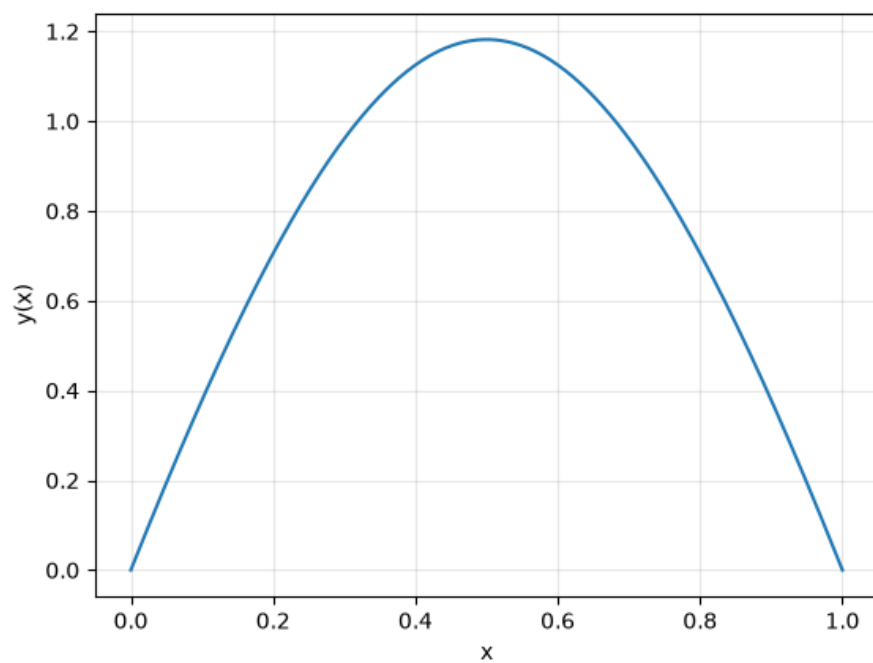


Figure 2: Solution $\lambda = 3.5135$.

$\lambda = 3.513830719$ — **exactly at the fold**

Here the run gets interesting.

```
INFO: single shooting found NO representable slope (every trial missed
      or diverged on one side) -- a corner-layer solution whose slope
      sits within machine epsilon of a blow-up wall evades shooting
      entirely. Collocation located the solution and sirpy VERIFIED it
      by substitution: nonexistence is RULED OUT.
```

```
INFO: single-shooting scan found no root; trying collocation before
      any nonexistence claim.
```

```
INFO: the problem was handed to SciPy's collocation solver. Solution
      by scipy.integrate.solve_bvp -- credit to the SciPy project.
      The result is a NUMERICAL grid + interpolant, not a power series.
```

Three things happened here, and each is worth naming.

It failed, and diagnosed why. At the fold the two solution branches have nearly merged — their initial slopes are 3.999971 and 4.000029, separated by six parts in a hundred thousand. Shooting must thread that needle. SIRPY did not report a generic failure; it identified the geometry that defeated it.

It refused to claim nonexistence prematurely. This is the discipline that makes the next section trustworthy. Failing to find a solution is not evidence that none exists, and SIRPY said so explicitly — trying collocation *before* any nonexistence claim.

It credited the tool it borrowed, and labelled the output. When SciPy's collocation solver found the solution, SIRPY said so by name, and marked the result as a numerical grid rather than a power series. It then verified that borrowed answer by substitution rather than accepting it.

The subsequent partition solve returned:

```
gamma_1 = 3.999971362
verification ... to 1.66e-12 (max RELATIVE residual 1.79e-13),
boundary data to 5.55e-17
```

The exact lower-branch slope at λ_c is **3.999971**. SIRPY returned **3.999971362**.

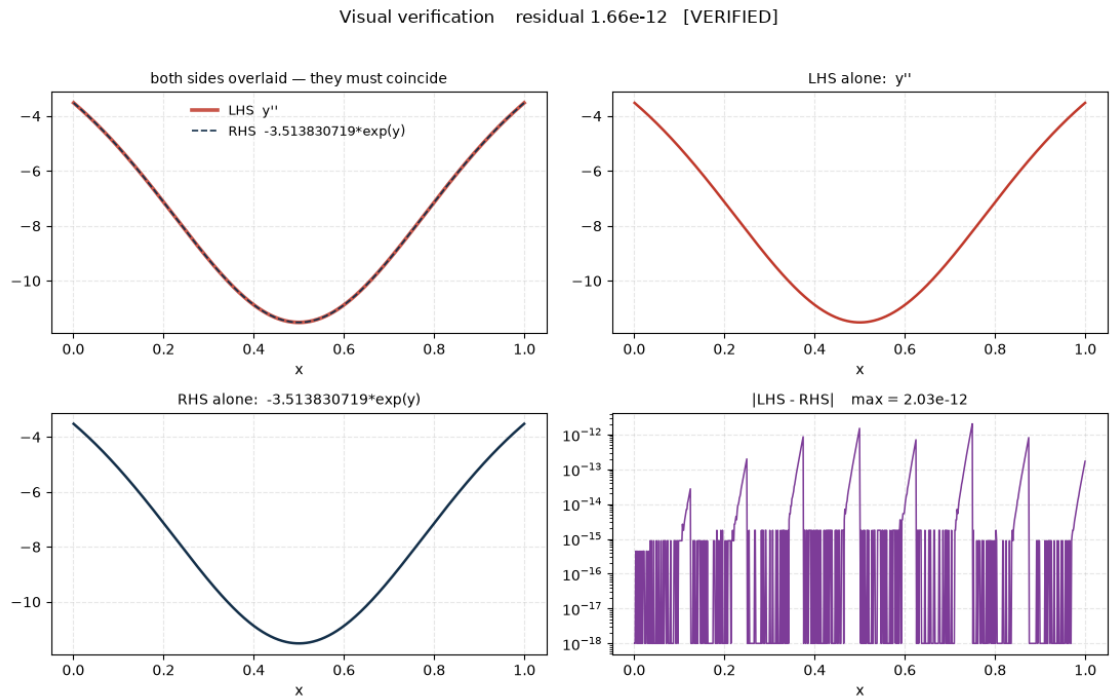
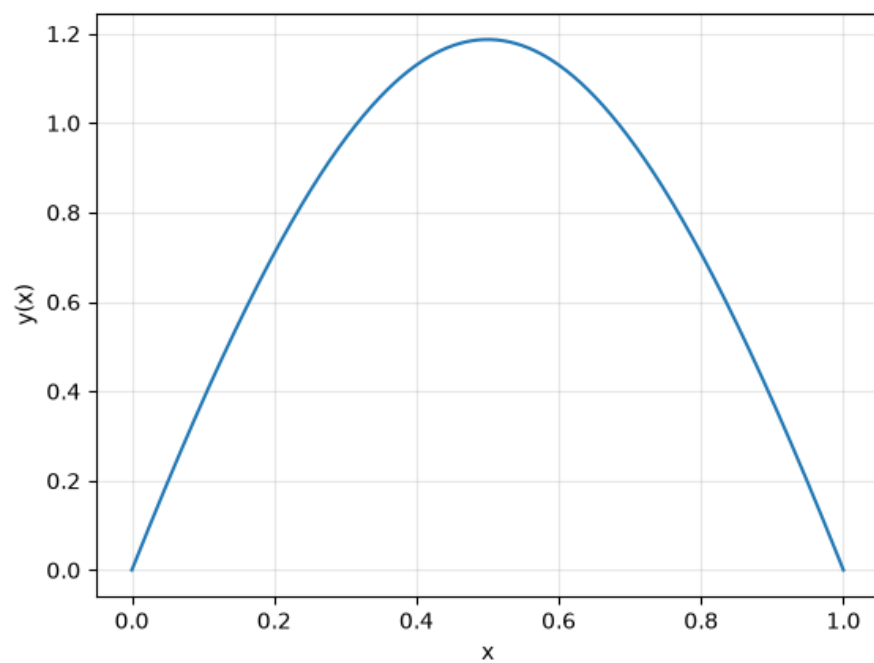


Figure 3: Verification $\lambda = 3.513830719$: both sides of the equation compared across the interval.



$\lambda = 3.52$ — **past critical**

CERTIFICATE OF NONEXISTENCE -- no slope satisfies these boundary conditions.

INFO: A scan of 35 trial slopes gamma in [-32, 32] gives a terminal mismatch $y(1) - 0$ that is NEGATIVE for every trial: the mismatch never changes sign, so no root exists in the scanned range.

INFO: Smallest achievable |mismatch| = 3.593e-03 at gamma = 3.967314; that floor trajectory is itself a verified solution of the ODE (max relative defect 8.3e-11) -- it simply cannot meet $y(1) = 0$.

INFO: Solutions of this two-point problem exist only up to $x_1 = b^* \sim 0.999123$: a saddle-node fold where two solution branches merge. Your $x_1 = 1$ lies 0.000877 beyond the fold. Any $x_1 \leq b^*$ is solvable, with TWO branches.

INFO: Certificate scope: single shooting unknown, scanned range as above (about 4.5x the natural slope scale); a root outside that range would evade the scan. This is a measured finding, not a raised error.

This is the entry. Not a failure message — a structured argument that no solution exists, with the evidence attached.

It states the scan performed and its range. It reports the closest approach achievable — a mismatch floor of 3.6×10^{-3} — and notes that this near-miss trajectory *is itself a verified solution of the differential equation* that simply cannot meet the second boundary condition. It identifies the obstruction by name: a saddle-node fold. And it locates that fold quantitatively, in terms of how far the interval could have been shortened to make the problem solvable.

Then, unprompted, it states **the limits of its own certificate**: the scan covered a specific range, and a root outside that range would have escaped it. A claim of nonexistence with its own scope attached.

Visual verification residual 3.55e-15

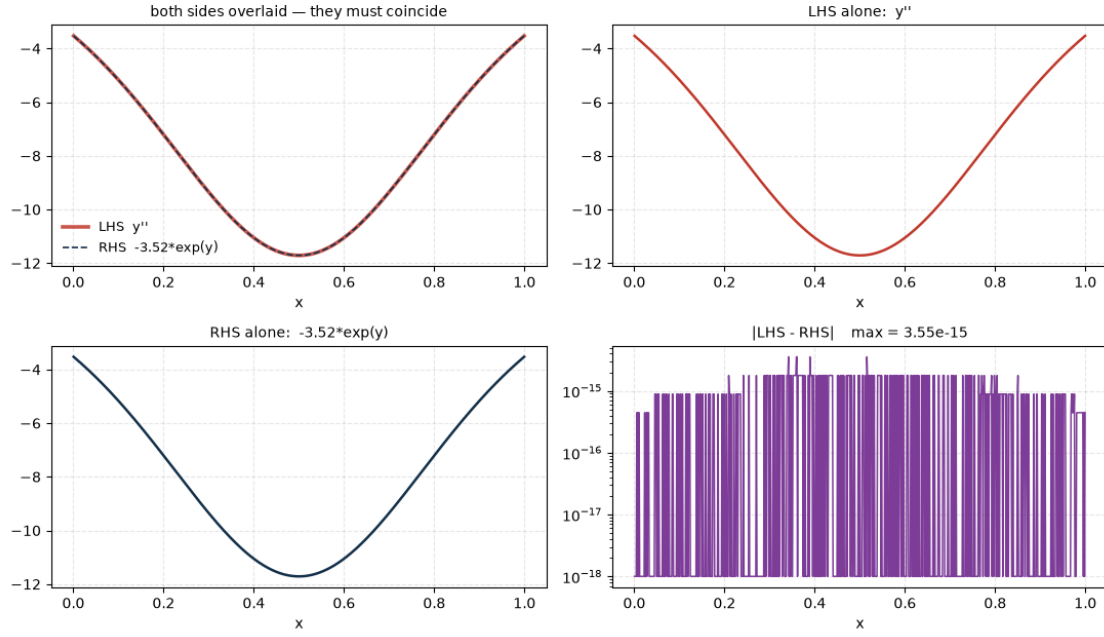


Figure 4: Verification $\lambda = 3.52$: both sides of the equation compared across the interval.

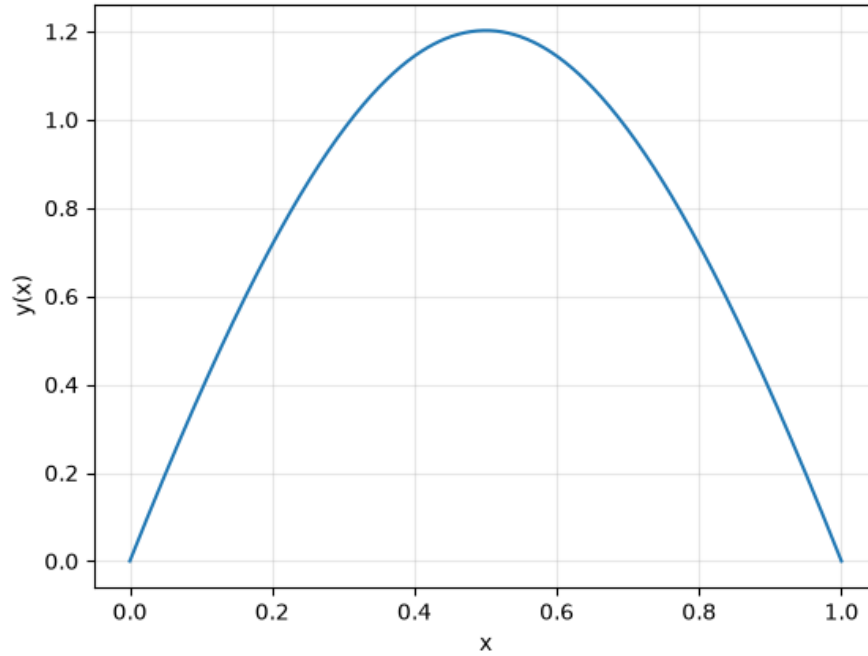


Figure 5: Solution $\lambda = 3.52$.

Checked against the closed form

Bratu has an exact solution, so every claim above can be checked from outside SIRPY:

	SIRPY	exact
Critical λ	(never told)	3.513830719125
Slope at $\lambda = 3.513830719$	3.999971362	3.999971
Fold location b^* for $\lambda = 3.52$	0.999123	0.999123297
$\lambda = 3.52$	no solution	no solution

The nonexistence claim is correct. And the fold location — reframed by SIRPY as an interval length rather than a critical λ — matches the exact value to six figures.

It refused to draw what doesn't exist

Asked to plot the $\lambda = 3.52$ result:

NONEXISTENCE CERTIFICATE: no solution curve exists to plot.

The numbers are in `r['certificate']` and the explanation in `r['info']`; returning `None`.

A solver that produced a plausible-looking curve here would be worse than useless. This one declines, and says where the evidence is instead.

What this problem shows

→ **Below critical, solved and verified** — relative residuals near 10^{-11} , boundary data at 10^{-16} , interface continuity measured.

→ **At the fold, it failed and explained the geometry** — two branches separated by 6×10^{-5} , a needle shooting cannot thread.

→ **It did not confuse failure with nonexistence**. Collocation was tried *before* any such claim was made. That discipline is what makes the next point credible.

→ **Past critical, it certified nonexistence** — with the scan, the mismatch floor, the fold location, and the scope of its own argument all stated.

→ **Every claim checks out against the exact solution**, including the fold location to six figures.

→ **It credited borrowed work and labelled it**. When SciPy solved it, SIRPY said so, and marked the output as numerical rather than analytic.

A note on scope

The certificate is explicit about what it covers: a scan of a stated slope range, about $4.5\times$ the natural scale of the problem. SIRPY says plainly that a root outside that range would have evaded it. We report it the same way — as a measured finding with stated limits, which is what it is. ## The problem library

This entry is part of the SIRPY Problem Library — an ongoing, public catalog of mathematical systems solved and verified by SIRPY.

- **GitHub** — github.com/Mitra-Institute-of-Education-MIE/sirpy-problem-library
- **Web** — mitrainstituteofeducation.org/sirpy

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Contact: info@mitrainstituteofeducation.org · mitrainstituteofeducation.org/contact

Please only send problems you are free to share publicly.

SIRPY is in final validation prior to its first public release. Every line quoted above is verbatim from the solver's output; nothing is reproduced from memory.

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